uc3m | Universidad Carlos III de Madrid

University Degree in Telematics Engineering
2024-2025

Bachelor Thesis

“Machine Learning-Based Predictive Modeling of Energy Prices”

Rodrigo De Lama Fernández

Emilio Parrado Hernández

Madrid, España, May 31st 2025

SUMMARY

This project focuses on the development of a machine learning-based predictive model for electricity prices in Spain. Using historical data from the OMIE and technical analysis (TA) indicators, the model aims to accurately forecast hourly energy prices. The study focuses on a single hourly slot, evaluating the performance of various Machine Learning models, such as linear regression, Lasso, and Random Forest. There was a strong focus on the feature engineering, employing technical analysis indicators such as moving averages, exponential moving averages, and momentum metrics. The results display the impact of tailoring the features to improve model accuracy and offer insights into the potential of data-driven approaches for energy price forecasting.

Keywords:

Energy

Machine Learning

Sliding Window

Technical Analysis (TA)

DEDICATION

Dedicated to my late grandfather who was not able to see me become an engineer like himself
Here is to my amazing family
To my parents, that have always helped me push through hard moments To my grandparents
To my girlfriend
To my friends and colleagues
To anyone and everyone that has supported me during my years in university

Here is to the next steps in life

CONTENTS

LIST OF FIGURES

LIST OF TABLES

1. INTRODUCTION TO THE PROBLEM

- **1. Explanation of how the price of energy works in Spain**
  - a. Energy
  - b. Prices
- **2. Machine Learning - Lets run through some model overviews, which are available to choose, but without actually referencing anything**

Energy has increasingly become more important. Availability is key for a modern high functioning society In Europe we are lucky to have a really advanced and reliable network. It is a key resource needed to succeed in the ecological transition that we are currently undergoing world wide. Electric mobility is a key element of this process. Electric cars, buses and trucks, need easily available and convenient sources of energy to recharge, to be able to play their role in society. An important factor in this ever growing necessity is the price of the available energy. But in recent times, that easy and affordable access has changed, and has become more unpredictable. Ever since the commencement of the war between Russia and Ukraine marked and inflection point in the mostly cyclical nature of electricity prices. Prices have gone up considerable since a couple years time, and it has been affecting everyone, from big consumers such as companies, to everyday people, having to pay a more expensive electricity bill at the end of the month. This has become a topic of great concern, with more and more people thinking about it often.

This led me to think about pricing, how it was structured, and what to expect as an end consumer. Could we plan our use of energy according to the price? Could it be predicted accurately? These were some of the questions I had that sparked my interest in looking into energy predictions.

In the initial discussions of this project, we talked about various ways to attack the problem at hand. We thought about a variety of systems but settled on a plan to explore the predictions with the more established methods of Machine Learning. This was done specially since it was a natural continuation of what was learned in the third year subject of Telematics Engineering, Modern Theory of Detection and Estimation.

The first idea we developed was to separate out the project in 24 blocks, one for each hour block to predict. The reason behind splitting the data in 24 blocks was simple, we assumed that data between the same time slot across neighbor days, would be similar, or would follow a certain trend.

Focusing on the innovation side of things, we decided to simplify the problem. Pivoting from creating an algorithm to predict hourly prices, to splitting that into the 24 blocks, and predicting a single one. We decided on the 14th hour of each day, since irrespective of the day of the week, or holiday, it would always be a valley period.

This kind of granular breaking up of data is similar to what a Random Forest does in order to absorb more details and create higher resolution predictions. We made this assumption based on the idea that it would simplify the project substantially.

Talk about Emilio's background in investment banking at BBVA and his alorithms.

Talk about the idea of using technical analysis for feature engineering

2. BACKGROUND AND RELATED WORK

2.1. General background

How is the price of electricity set in Spain and Portugal?

What happens when new producers enter the market and sell to the energy pool?

Comment about more typical case studies focusing on energy consumption, since that is something more consistent and predictable

2.2. The Sliding Window plan: Setting up the Feature Matrix

How do I prepare my data to utilize across different models: The Feature Matrix (weight matrix)

The sliding window theory (mini-models)

2.3. Which Machine Learning models will be used

What algorithms will I use?

- Linear Regression
- Lasso Regression
- Random Forest

2.4. Fine Tuning with Technical Analysis and Feature engineering

What Type of finetuning will I use?

- Feature engineering & selection
- Technical analysis based feature engineering

TA metrics used? SMA, EMA, ROC & RSI

Talk about the intervals used:

$SMA_3, SMA_5, SMA_7, SMA_{14}, SMA_{30}, SMA_{60}, SMA_{90}, SMA_{180}, SMA_{360}$

$EMA_3, EMA_5, EMA_7, EMA_{14}, EMA_{30}$

$ROC_3, ROC_5, ROC_7, ROC_{12}, ROC_{14}, ROC_{30}$

RSI_5, RSI_7, RSI_{14}

2.5. Evaluation Metrics

How will I measure my accuracy? errors: R^2 RMSE & MSE

3. THEORETICAL SYSTEM PROPOSAL DESCRIPTION

This chapter will outline the design of the system developed for the project, describing the complete pipeline constructed for the system. This will be explained starting from a blank state, detailing the complete set-up process required to run the project. Following this initial configuration, the data preparation stage will be outlined. This will be continued by the feature matrix creation process, concluding with the training of the machine learning models and the chosen evaluation methods.

The following Block diagram will model in the most basic forms the system design:

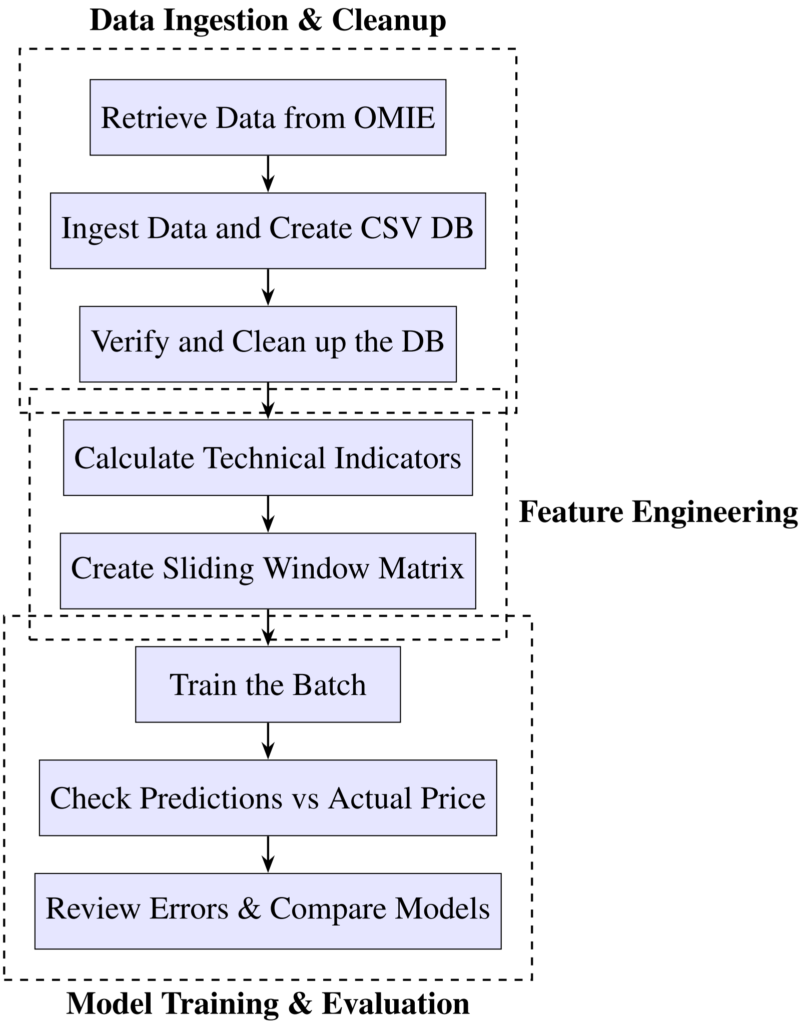


Fig. 3.1. Theoretical Block Diagram of the Project Pipeline

Data Preparation

Matrix Creation

Mini-models - sliding window approach

The thought process behind our sliding window approach has to do with the data and the No estacionario. No son datos IID - Independientes e identicamente distribuidos. Hay correlacion en mis datos entre cada dia? Si deberia ser relativamente alta, estacional, pero progresiva y lenta.

Technical indicators Que es cada cosa y porque usarlos en vez de series temporales, porque la info temporal esta implicita en los indicadores. Indicadores que capturan patrones en la serie Ventaja: modelo ML mas sencillo pq las features son muy informativas vs

ML blocks Linear Regression Linear combination of every feature Lasso Regression Can turn unimportant features into 0s Random Forest Feature engineering

Pandas, NumPy, scikit-learn

Retrieve Data from OMIE

Kick-starting the pipeline design, the data is retrieved from the OMIE website, and is ingested into a CSV file that will act as our rudimentary database. The process is done with the following code:

After a successful retrieval, the database is checked for missing or repeated data points. This will be checked in a time-slot basis.

Since the idea is to cater to any potential system, it was created without any hour slot preference, but in the system I will be proposing, we will limit the dataset to just the 14h time-slot.

The database will be transformed into a reduced set, containing only the relevant data points, measured always at 14h

Subsequently comes the creation of the feature matrix. This will initially consist of exclusively the sliding window width desired for that training batch.

Once the matrix is created (and verified), the training process begins.

Due to our particular system, the training will consist of complete rows of that feature matrix.

Cross validation?? Not really done, in another way w the sliding window cambio parametros pero no la arquitectura del modelo

Review the mini model theory, why is it reasonable to not use training set, validation and test sets?

Review error rates and compare with other iterations/ batches and other models.

Detailed Steps:

Prior set up:

Developed and tested with Python 3.13 [1]

To set up a new virtual environment [2] run:

```
python -m venv .venv
```

To activate the new virtual environment run:

```
in macOS
source .venv/bin/activate
in Windows
.venv\Scripts\Activate.ps1
```

Install the following contents to reproduce the original virtual environment:

```
# To install the following packages run:
# pip install -r requirements.txt
ipykernel==6.29.5
matplotlib==3.10.1
numpy==2.2.3
pandas==2.2.3
requests==2.32.3
scikit-learn==1.6.1
ta==0.11.0
```

1. Download of OMIE data (only section independent of the whole system) [3]

Downloaded with a Python script for the dates that were not available with an easy .zip file (for whole previous years):

Insert the python script?? better suited for an annex??

```
import requests
from datetime import datetime, timedelta

# Usage instructions:
    # Set file name to save the filenames to download
    # Set start and end dates

file_path = 'to_download.txt'
start_date = datetime(2025, 2, 14)
end_date = datetime(2025, 3, 18)

# Compose filenames to download
def compose_filenames():
    start_date = start_date
    end_date = end_date

    # Generate the entries for the number of days comprehended
```

```
    to_generate = (end_date - start_date).days + 1
    entries = []
    for i in range(0, to_generate):  # x days from start to end
        date_entry = start_date + timedelta(days=i)
        entries.append(f'marginalpdbc_{date_entry.strftime("%Y%m%d")}.1')

    # Save to a .txt file
    with open(file_path, 'w') as file:
        for entry in entries:
            file.write(entry + '\n')

# Example URI to build
# https://www.omie.es/es/file-download?parents%5B0%5D=marginalpdbc&filename=marginalpdbc_20230102.1
def read_filenames_and_compose_urls(file_path):
    base_url = "https://www.omie.es/es/file-download?parents%5B0%5D=marginalpdbc&filename="

    # Read the filenames from the .txt file
    with open(file_path, 'r') as file:
        filenames = [line.strip() for line in file if line.strip()]

    # Compose the URLs
    urls = [base_url + filename for filename in filenames]

    return urls

def download_files(urls):
    for url in urls:
        # Extract the filename from the URL
        filename = url.split('=')[-1]

        # Send a GET request to download the file
        # response = requests.get(url)
        response = requests.get(url)

        # Check if the request was successful and write to file
        if response.status_code == 200:
            # Write the content to a file with the extracted filename
            with open(filename, 'wb') as file:
                file.write(response.content)
            print(f"Downloaded {filename}")
        else:
            print(f"Failed to download {filename}")

# Execution
compose_filenames()
urls = read_filenames_and_compose_urls(file_path)
download_files(urls)
```

2. Data shape? NEXT CHAPTER 4

3. Data format: NEXT CHAPTER 4

4. Data organization

We will transform the data into a CSV format easier to use with the Pandas library to later create a DataFrame. We will maintain the same information, but in a encoding more suitable for Pandas. For that, the date and time must be transformed to a format such as Datetime which will transform the information from `2018;01;01;14; ...` to `2018-01-01 14:00:00`

5. Data visualization with Seaborn? With Microsoft Corporation's "Data Wrangler" VSCode extension

6. Data retention of only the relevant info time and date, and MarginalES - drop MarginalPT

7. Data cleanup: Check for missing or duplicate entries. since Daytime Savings introduces duplicate hours some days of the year: INSERT some dates in the dataset that have the duplicated entries

8. Explain all blocks of the system? Data download: Data ingest

Contar los bloques empleados en cada parte – como junto las piezas para construir el sistema – porque uso las piezas que uso

Hablar de implementaciones como que librerías, que scripts etc – no es muy necesario, mas importante la idea

4. EXPERIMENTATION

In this chapter we will review all of the results, and discuss the different set-up variations for the models.

4.1. Data Description – OMIE data

The data provided by the OMIE takes this shape:

```
MARGINALPDBC;
2018;01;01;1;28.1;6.74;
2018;01;01;2;33;4.74;
2018;01;01;3;32.9;3.66;
2018;01;01;4;28.1;2.3;
2018;01;01;5;27.6;2.3;
2018;01;01;6;24.6;2.06;
2018;01;01;7;20.1;2.06;
2018;01;01;8;19.9;2.06;
2018;01;01;9;19.84;2.3;
2018;01;01;10;19.9;2.3;
2018;01;01;11;19.9;2.3;
2018;01;01;12;19.9;2.3;
2018;01;01;13;23.6;2.3;
2018;01;01;14;25.1;2.3;
2018;01;01;15;23.6;5;
2018;01;01;16;24.6;5;
2018;01;01;17;25.1;5;
2018;01;01;18;27.6;8.85;
2018;01;01;19;27.6;15.93;
2018;01;01;20;28.1;22.02;
2018;01;01;21;32.9;20;
2018;01;01;22;28.1;21.95;
2018;01;01;23;28.1;23.52;
2018;01;01;24;27.6;16.35;
*
```

We can interpret this format following the OMIE's guide: *Modelo de Ficheros para la distribución pública de Información del mercado de electricidad 1.35*[4]. In page number 67, chapter 6.18 we may find the following information regarding the encoding format for the information:

```
6.18 Precios marginales del mercado diario (MARGINALPDBC)

Fichero con los precios marginales del Mercado Diario para cada una
de las horas.
Nombre del fichero: marginalpdbc_aaaammdd.v donde aaaammdd
corresponde a la fecha de sesión y v es la versión del fichero.

Descripción de los campos:
CAMPO DESCRIPCIÓN VALORES VÁLIDOS
Año Año I4 - 20XX
Mes Mes I2 - 1 a 12
Día Día I2 - 1 a 31
Hora Hora I2 - 1 a 25
MarginalPT Precio marginal zona Portuguesa F8.2 -
-99999.99 a 99999.99
MarginalES Precio marginal zona Española F8.2 -
-99999.99 a 99999.99
```

4.2. Set Up Explanation

How do I use the previously explained system - what parameters did I modify to obtain the final results. Feature engineering, Alpha variation testing in Lasso, Tree depth and number of leafs. Que modelos y que configs

4.3. Results of the testing

Results as graph/ tables - This would be to compare a model across various iterations. Model best-run comparisons, etc.

Error medio en todas las predicciones Y percentiles (expectation shortfall tambien?)

4.4. Low level discussion

This would be the final explanation that I would give to a colleague of mine, with full details on how I have done everything

5. REGULATORY FRAMEWORK

Not essential for this investigative project because we do not go to market.

5.1. Data Availability

Habria que hablar de las normas, pq si alguien lo usa es para ir a mercado

Current Spanish and European regulations allow for the use of publicly available institutional data for use in educational investigative work.

I need to talk about laws, what data is permissible to use and how. This would be essential for a project that does indeed go to market.

For us, its not that relevant.

5.2. Software & Licenses

For the development of this project I have used a variety of open and closed source utilies and code.

For the sake of organization, I will categorize this section into two distinct parts, required software to reproduce the project, and optional software for ease of production, plannification, organization or otherwise related.

Required:

Python - This was the programming language in wich the entirety of the project was developed [1]

Scikit-learn - library for ml [5]

Pandas - data management [6]

ta library - Technical Analysis library[7]

Only light use of Seaborn - data visualization [8]

NumPy - number manipulation [9]

macOS / Windows - closed source platforms across where the project was developed [10] & [11]

Optional:

Git - a distributed version control system [12]

GitHub - Repository hosting platform [13]

Visual Studio Code - open source text editor for my general coding needs [14]

Visual Studio Code extensions such as Microsoft Data Wrangler, Python packages or such

Python venv - This was created as it is best practice to create new Python virtual environments for each new project, as specially in a production environment these usually require certain specific versions that must be met in order for all components to work as intended. An alternative method of dealing with this would be using Conda, and creating with it a new Conda environment, [2]

Overleaf - closed source platform used to compile LaTeX code [15]

LaTeX - open source language used to create the final project document [16]

OpenAI ChatGPT, Google Gemini & Anthropic Claude - closed source large language models used to aid in troubleshooting general coding issues

6. SOCIO-ECONOMIC ENVIRONMENT

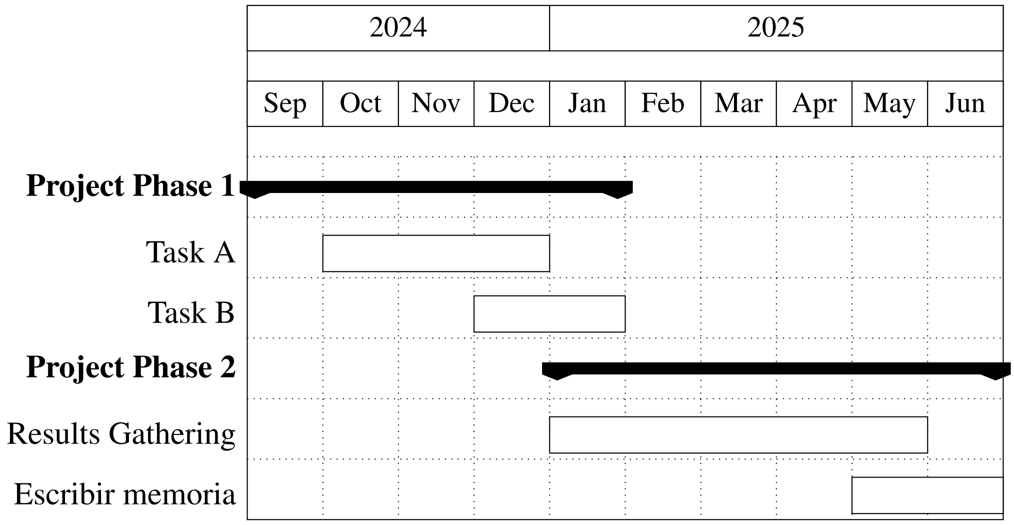
6.1. Socio-economic Impact

Why is this relevant?

- Energy Prediction - Shutdowns - Price sensitivity

6.2. Project plan

Gantt diagram representing time spent - Design and innovation time frames.



6.3. Budget

The complete price breakdown of the investigation is outlined in the following tables:

Tool	Price (€)	Amortization per year (€)	Useful Life (years)	Usage Time (months)	Cost (€)
MacBook Pro M1 Pro	1250	312.5	4	9.5	247.4

TABLE 6.1. TOOL AMORTIZATION

Hourly rates table

Expert Machine Learning consultant (Emilio, PhD)

Junior engineer (pre-graduate)

Name	**Role**	**Price (€/hour)**	**Hours**	**Cost (€)**
Emilio Parrado Hernández	Expert ML Consultant (PhD)	200	-	-
Rodrigo De Lama Fernández	Junior Engineer (Pre-Graduate)	30	-	-

TABLE 6.2. HUMAN COSTS

Concept	**Cost (€)**
Direct/Material Costs	247.4
Engineering Costs	-
University Carlos III Costs - Expert ML Consultant	-
Total	**-**

TABLE 6.3. GENERAL COST BREAKDOWN

7. CONCLUSIONS

In this final chapter we will review the general conclusions attained throughout the development of this project's machine learning based solution for energy price predictions. It will also discuss potential development paths for future work pertaining the system designed in this investigative project

7.1. Recap of the project

After finishing the whole project, read it, and reintroduce the objectives to the reader. Remind them of the completion of them

7.2. Revisit the objectives

Predicting the prices in an accurate manner using Machine Learning algorithms and technical analysis

7.3. Future work

Less error?

Less

BIBLIOGRAPHY

[1] Python Software Foundation, *Python programming language*, https://www.python.org/about/, 2025.

[2] Python Software Foundation, *Venv — creation of virtual environments*, https://docs.python.org/3/library/venv.html, 2025.

[3] OMIE, *Precios horarios del mercado diario en españa*, https://www.omie.es/es/file-access-list?parents=/Mercado%20Diario/1.%20Precios&dir=Precios%20horarios%20del%20mercado%20diario%20en%20EspaÃśa&realdir=marginalpdbc, Accessed on June 3, 2025, Jun. 2025.

[4] OMIE, *Modelo de ficheros para la distribución pública de información del mercado de electricidad 1.35*, https://www.omie.es/sites/default/files/2024-06/Formato_ficheros_inf_pub_135_0.pdf, Accessed on June 3, 2025, Jun. 2024.

[5] The scikit-learn Development Team, *Scikit-learn: Machine learning in python*, https://scikit-learn.org/stable/, 2025.

[6] The Pandas Development Team, *Pandas: Python data analysis library*, https://pandas.pydata.org, 2025.

[7] bukosabino, *Technical analysis library in python (ta)*, https://github.com/bukosabino/ta, 2025.

[8] Michael Waskom et al., *Seaborn: Statistical data visualization*, https://seaborn.pydata.org, Used lightly for visualization, 2025.

[9] The NumPy Development Team, *Numpy: Array programming for scientific computing*, https://numpy.org, 2025.

[10] Apple Inc., *Macos sequoia*, https://www.apple.com/macos/macos-sequoia/, 2025.

[11] Microsoft Corporation, *Windows 11*, https://www.microsoft.com/windows/windows-11, 2025.

[12] Git, *Git: A distributed version control system*, https://git-scm.com, 2025.

[13] GitHub, *Github: Repository hosting platform*, https://github.com/about, 2025.

[14] Microsoft Corporation, *Visual studio code*, https://code.visualstudio.com/docs/editor/whyvscode, Open source code editor, 2025.

[15] Overleaf, *Overleaf: Online LaTeX editor*, https://www.overleaf.com/about, 2025.

[16] LaTeX Project, *Latex: A document preparation system*, https://www.latex-project.org, 2025.